Orientation Detector for Ship Targets in SAR Images Based on Semantic Flow Feature Alignment and Gaussian Label Matching

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***Abstract*— To address the challenges in synthetic aperture radar (SAR) ship target detection, this article proposes an SAR ship small target orientation detector named FADet based on semantic flow feature alignment and Gaussian label matching. First, to solve the feature misalignment problem caused by feature extraction downsampling and residual connections, we introduce the FAM module into FPN, which automatically aligns deep and shallow fine-grained semantics information through semantic flow alignment. Second, due to the scattering characteristics of SAR imaging, the boundary information of SAR targets is not obvious, we combining attention mechanisms design an adaptive boundary enhancement module to enhance the target boundary information. Finally, to solve the problem that small targets have difficulty matching positive samples under IoU rules, we design a label matching strategy based on Gaussian distribution. This matching strategy can still learn regression information when two boxes do not intersect. Based on the SSDD+ and RSDD-SAR datasets, the effectiveness of each module in FADet is verified by ablation experiments. Additionally, through comparison experiments with the latest orientation detection methods, FADet achieves a good compromise between accuracy and inference speed. The AP50 and AP75 on the SSDD+ and RSDD-SAR is 91.03, 59.94 and 90.78, 59.91, respectively, and the FPS is 19.83.**

***Index Terms*— Adaptive boundary enhancement, deep learning, Gaussian label matching, oriented synthetic aperture radar (SAR) object detection, semantic flow alignment.**

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I. Introduction

As an active imaging system, synthetic aperture radar (SAR) has the ability of all-weather and all-day imaging, so it is widely used in civil and military fields [1]. Ship detection is an important task for the intelligent interpretation of SAR images and is of great importance in the supervision of commercial ships on the sea and the monitoring of military activities. SAR ship detection aims to locate ship targets from complex offshore sea clutter and inshore built-up backgrounds. In recent years, with the development of SAR imaging technology, high-resolution SAR ship data have become easier to obtain, and research on SAR ship detection has gradually increased.

SAR target detection has gone through a development process from traditional methods to deep learning methods. The most classic traditional SAR target detection algorithm is the constant false-alarm rate (CFAR). CFAR dynamically adjusts the detection threshold according to radar clutter data to maximize the target detection probability while the false alarm probability remains unchanged. Many detection algorithms have been proposed based on the CFAR. Within the framework of a multiplication model, Gao and Shi [2] used the reciprocal of the gamma distribution to describe the texture components of sea clutter in a nonuniform background and proposed a CFAR detection model for nonuniform sea clutter. Li et al. [3] used complex signal kurtosis (CSK) as a statistical index to determine the truncation depth to reduce outlier pollution in clutter samples, improving the accuracy of clutter modeling. Li et al. [4] proposed an adaptive superpixel-level CFAR method using unsupervised clustering and a nonlocal topology strategy. The algorithm based on CFAR depends on the clutter statistical distribution. When the background is complex, the CFAR detection performance drops sharply, so the CFAR detection performance for inshore targets is limited.

Due to the strong feature extraction ability of convolutional neural networks (CNNs), object detection based on CNNs has developed rapidly in recent years. In the field of optical images, scholars have proposed many algorithms based on CNNs. These algorithms can be roughly divided into two categories: one stage [5], [6], [7], [8], [9] and two stage [10], [11], [12]. With the success of CNN-based algorithms in the optics field, researchers have begun to apply these algorithms to the SAR image field. In recent years, combined with the characteristics of SAR images, many SAR ship

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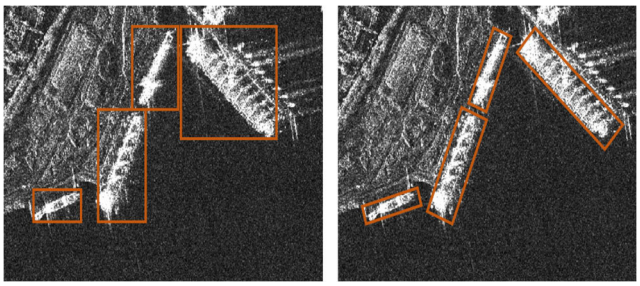


Fig. 1. (Left) Horizontal box detection, which contains lots of background. (Right) Oriented detection with less background.

detection algorithms based on CNNs have been proposed [13], [14], [56]. Zhu et al. [15] proposed a high-speed and high-precision detector (H2Det) using a CSP module, spatial pyramid pooling, bottom-up path enhancement, and mosaic data enhancement. Gao et al. [16] proposed a fast and efficient method based on YOLOv4, aiming to improve the detection performance of small targets. Hu et al. [17] proposed a BANet detector based on an anchor-free architecture that constructs a local attention module (LAM) and nonlocal attention module (NLAM) through deformable convolution to extract local and nonlocal features of the target to improve the target detection performance.

Although the algorithm based on horizontal bounding box (HBB) has made certain progress in the field of SAR ship detection. Due to the diversity of SAR imaging perspectives, ship targets in SAR images typically appear in any direction, are arranged densely, and have a large aspect ratio; these issues limit the detection performance of HBB, which often contains considerable background clutter, as shown in Fig. 1. This easily causes classification and regression errors. As a result, researchers have developed various algorithms based on oriented bounding boxes (OBBs). The OBB algorithms add angle regression on the basis of HBB to make the detector locate the target more accurately. In the optical remote sensing image field, researchers have developed a series of algorithms, including RoITransformer [18], R3Det [19], ReDet [20], and oriented R-CNN [21]. Based on these algorithms, scholars have developed many SAR image detection algorithms [57], but these algorithms do not fully consider the characteristics of SAR images. First, these algorithms do not consider the semantic feature misalignment problem introduced by feature extraction network downsampling and residual connection operations. Second, compared with objects in optical images, ship targets in SAR remote sensing images are relatively small, and the uniqueness of SAR imaging system results in low image contrast, loss of target texture details, and indistinct target boundaries. Third, affected by scattering, the visual separability between the target and the background is poor. Therefore, in response to the above issues this article proposes an SAR ship small target orientation detector called FADet based on semantic flow feature alignment and Gaussian label matching. The main contributions of this article are as follows:

1) First, the downsampling and residual connection operations in the CNN lead to the dislocation of different resolutions features, which affects the detection performance. To better transmit semantic information from shallow features to deep features, we design a semantic flow-based alignment module. Inspired by optical flow, semantic flow using different resolutions features as input and evaluates and aligns the feature mapping between the two feature maps through flow.
2) Second, due to the scattering mechanism of SAR imaging, the texture features of SAR targets are not obvious, with the downsampling operation of convolution, and the boundary information of the target is gradually lost, especially for small targets. To further strengthen the target features, we have designed a hybrid attention module that enhances target features by combining channel and spatial attention. Then, the attention enhances feature addition of preserving target boundary information to expand the boundary information of the target and improve the target location performance.
3) Furthermore, we propose a label allocation strategy based on a Gaussian distribution to resolve the problem that small targets are difficult to match to positive samples based on the IoU matching strategy, which easily leads to small targets missing detection. Label matching based on a Gaussian distribution can still provide learning information when the two boxes do not overlap.

II. Related Works

Our method is mainly related to the oriented two-stage anchor box method, which involves feature alignment, attention mechanism, and label-matching techniques. To stress our contributions to these methods, they are introduced in this section.

A. Oriented Bounding Box (OBB) Target Detection

Researchers have developed many OBB detection algorithms based on optical remote sensing images [30], [31], [32], [33], [34], [36]. The two-stage oriented target detector based on anchor has better detection performance compared to the one-stage detector by coarse prediction and regression of anchor in the first stage and fine prediction and regression of proposal in the second stage. The two-stage oriented detector can be mainly divided into two groups. The first group generates a horizontal proposal in the first stage and an oriented target in the second stage. Typical methods include rotating RCNN and ReDet [20]. The second group generates oriented proposals through regression in the first stage, and refined proposals through further regression in the second stage. Typical methods include oriented RCNN [21] and DODet [30]. Oriented RCNN generates oriented proposals through the first stage midpoint drift representation, and the second stage further regresses the first stage's oriented proposals through regression angles. In the first stage of DODet, an OPN network was designed to generate oriented proposals to address the spatial mismatch between horizontal proposals and oriented objects. In the second stage, a localization guided detector (LDH) was developed to further address the problem of feature misalignment between classification and localization. In SAR ship images, the length and width ratio of the target is large, and only the angle is predicted in the second stage, which will easily cause the target and detection boxes to be misaligned. Therefore, the second group of methods is used in this article.

B. Feature Alignment

Feature alignment is currently divided into two groups. One group is the mismatch between the target and the detection

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box, and the other group is the mismatch between feature maps of different resolutions when extracting target feature maps. The first group of most typical algorithms, such as R3det [19], designs a feature refinement model based on feature interpolation, specifically overlaying feature maps through dual convolution to obtain new features, thereby obtaining precise regression of the target. S2A-Net [47] designs a feature alignment module that uses anchor refinement network ARN to generate a high-quality rotated anchor to match the regression of the anchor with the features between the targets. Another classic method is FaPN [48]. FaPN proposes a feature alignment module that learns the high-level features sampled on the alignment of the pixel's transformation offset. Zhu et al. [49] proposed a spectral nonlocal block in unifying nonlocal blocks for neural networks, which corrects the position shift between pixels through remote dependencies and further obtains aligned features during feature addition. The feature alignment module FAM designed in this article belongs to the second group of methods, which further enhances detection performance by aligning features between feature maps.

C. Attention Mechanism

The convolution neural network has different important contributions to the classification and detection of different channel features. The spatial position response on the feature map represents the feature value of the target. The attention mechanism aims to highlight the values of channels or spaces by differential weighting them. Researchers have proposed many attention mechanisms.

Attention mechanism also has important applications in SAR target detection; Chen et al. [35] combined with nonlocal attention proposed an R2FA-Det rotation detector to solve the problem of SAR image targets with a large aspect ratio, ambient direction, and dense distribution. Ma et al. [29] proposed an anchor-free SAR ship detection method based on key point estimation and an attention mechanism to solve the problems of complex background interference.

The attention mechanisms in SAR-oriented target detectors are mostly a single channel or spatial attention, and most of the attention mechanisms are inserted into the backbone or FPN. Our design uses an attention mechanism that combines channel and spatial, and innovative applies it in the proposal to enhance the features of the target.

D. Label Matching Strategy

Currently, both one-stage and two-stage anchor-based detectors mostly use IoU rules to match positive samples and negative samples. Based on IoU rules, researchers have developed GIoU, DIoU, CIoU, and other rules. These matching methods calculate the intersection of the union between the target and anchor. The IoU between the small target and the anchor is relatively small, and it is easy to lose the small target when the network training. Therefore, researchers have proposed many matching rules based on other rules, Wuhan University proposed the normalized Wasserstein distance [39], which solves the problem of IoU sensitivity to position deviation by calculating the Wasserstein distance between the anchor and the GT, greatly promoting the performance of small target detection. Huang et al. [50] proposed a general Gaussian heatmap label assignment. The adaptive Gaussian heatmap label assignment strategy is based on 2-D oriented Gaussian heat maps, simplifying the position definitions of positive and negative samples without any prior anchor boxes, and make the distribution of positive sample positions more flexible to adapt to the size and direction of the object. Xu et al. [51] proposed a receptive field-based label assignment (RFLA) strategy. RFLA first uses the prior information that the characteristic receptive field obeys the Gaussian distribution, and proposes a new receptive field distance (RFD) to directly measure the similarity between the Gaussian receptive field and GT, instead of using IoU or central sampling strategy to allocate samples.

III. Methods

In this section, we first introduce the overall structure of FADet. Then, FPN based on semantic flow design is introduced. Next, the adaptive boundary enhancement module is presented. By combining the attention mechanism and enhancing the boundary information of the target, the feature extraction ability of the target is enhanced, and the target positioning accuracy is increased. Finally, label allocation based on a Gaussian distribution is introduced to solve the problem that the anchor and GT do not overlap, which leads to small targets that cannot match the positive samples, resulting in missing detection.

A. Overall Architecture

The FADet algorithm is shown in Fig. 2. ResNet50 is used as the backbone to extract image features. Due to downsampling and residual connection operations in the backbone, the extracted shallow and deep layers have the problem of feature dislocation. To solve this problem, when features are input to the FPN, we use a semantic flow alignment module to align and add features of two different levels through semantic flow. After feature alignment, multiscale horizontal anchors are generated on the image. Then, a label allocation strategy based on a Gaussian distribution is used to allocate four positive samples to ground truth boxes. For regression of the anchor, the midpoint offset representation is used to generate the oriented proposal, and rotated RoIAlign maps the proposal to the size of 7*7. Furthermore, the boundary information of the proposal is enhanced through the attention mechanism to increase the positioning accuracy of the target. In the second stage, the fully connected layer is used to classify targets through decoupling heads, and the convolution layer is used to precisely regress the target by predicting the center point coordinates, length, width, and angle of the target.

B. FAM-FPN

The idea of semantic flow alignment originates from optical flow [38]. Optical flow utilizes the temporal changes of pixels in an image sequence and the correlation between adjacent frames to find the corresponding relationship between the previous frame and the current frame, thereby calculating the motion information of objects between adjacent frames. Analogous to optical flow, the relationship between two adjacent feature maps in the same image can also be represented by the "motion" of each pixel from one feature map to another. When the detection method based on deep learning extracts target features with downsampling and residual connections, there may be a misalignment of target positions in shallow and deep features, this process can be simulated as a motion between

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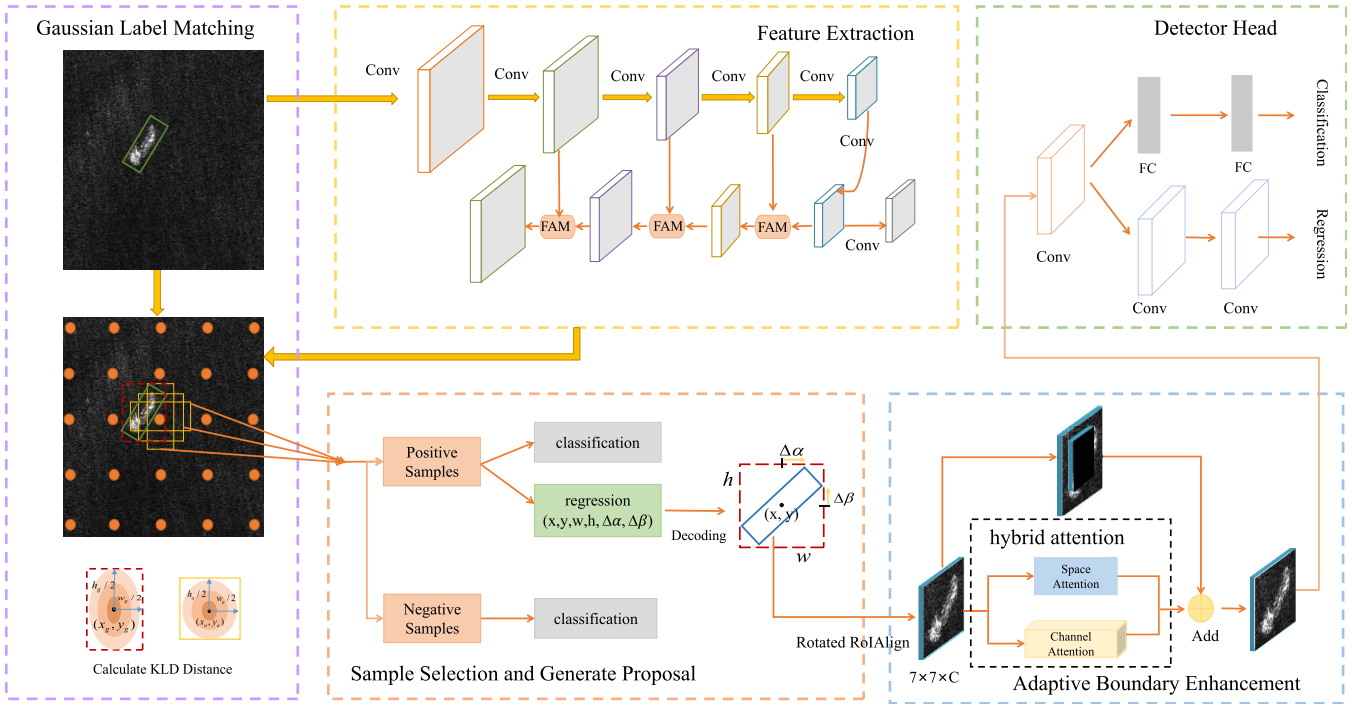


Fig. 2. Overall architecture of FADet network.

two features. Combined with FPN, semantic flow using feature maps at different resolutions as input and evaluating their internal differences to find suitable flow fields, dynamically indicating how to effectively align these two feature maps. The strategy used in the classic FPN two-layer feature fusion is to compress the number of channels of the shallow and deep features to the same dimension through a 1×1 convolution. Then, the deep features through the bilinear upsampling operation expand the feature resolution to the same size as the shallow features bilinear interpolation of a group of uniform sampling positions is used to restore the resolution of the down sampling feature map. Bilinear upsampling operation cannot effectively align misalignment features. As shown in Fig. 3, given two adjacent feature maps F_l and F_{l-1}, FAM first convert the feature map F_l and F_{l-1} to the same size. Then concatenate the two feature maps and used 3*3 convolution to predict the semantic flow field. This process can be described by the following formula:

$$\Delta_{l-1} = \text{conv}_l(\text{cat}(F_l, F_{l-1})), \quad \Delta_{l-1} \in \mathbb{R}^{H_{l-1} \times W_{l-1} \times 2}. \tag{1}$$

For each spatial position p_{l-1} in the semantic flow field Δ_{l-1}, it is mapped to a point p_l on the upper layer feature through upsampling and simple addition operation. This formula is as follows:

$$p_l = \frac{p_{l-1} + \Delta_{l-1}(p_{l-1})}{2}. \tag{2}$$

Then, using the differentiable bilinear sampling mechanism, interpolate the four points on p_l. Its mathematical formula can be described as:

$$\tilde{F}_l(p_{l-1}) = F_l(p_l) = \sum_{p \in N(p_l)} w_p F_l(p) \tag{3}$$

where $N(p_l)$ represents the neighboring point of warp point p_l in F_l, and w_p represents the bilinear kernel weight value.

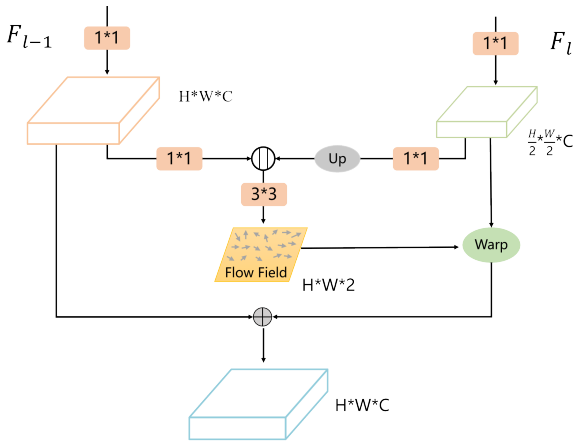


Fig. 3. FAM module.

From the Fig. 4, it can be seen that the feature map obtained through the FAM module (b) is more pronounced than the bilinear upsampling (a) feature.

C. Adaptive Boundary Enhancement Module

After extracting the features of the target, the feature of the target center point is relatively prominent. This is conducive to target classification, and the feature around the target boundary is more conducive to target regression. With the continuous convolution and downsampling operation, the boundary features of the target are lost. Therefore, we propose an adaptive boundary enhancement module to enhance the salient information of the target boundary to increase the positioning accuracy.

The proposal generated by RPN is mapped to a size of 7×7 through Rotated ROIAlign. In order to further highlight the target features, as shown in Fig. 5, we have designed a

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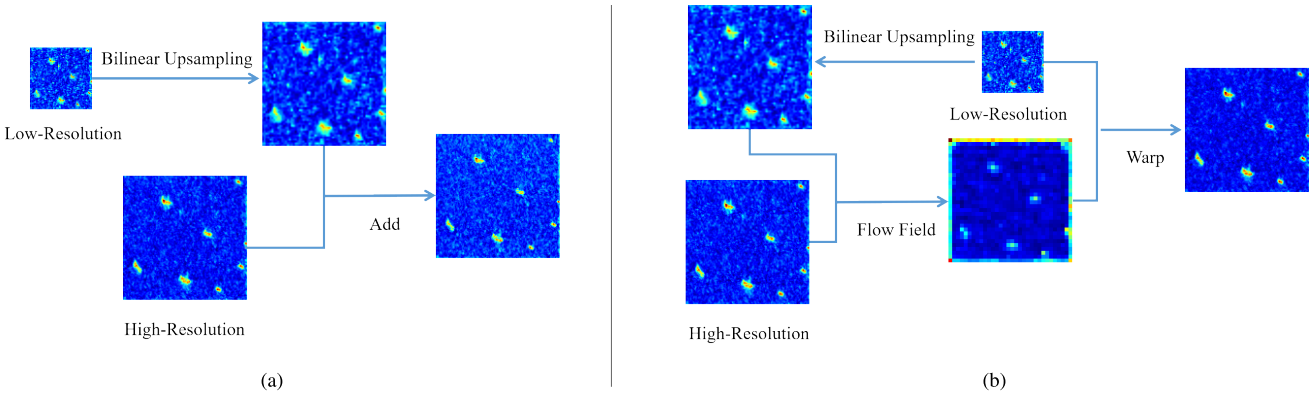


Fig. 4. Visualization of feature maps. (a) FPN visualization result. (b) FAM-FPN visualization result.

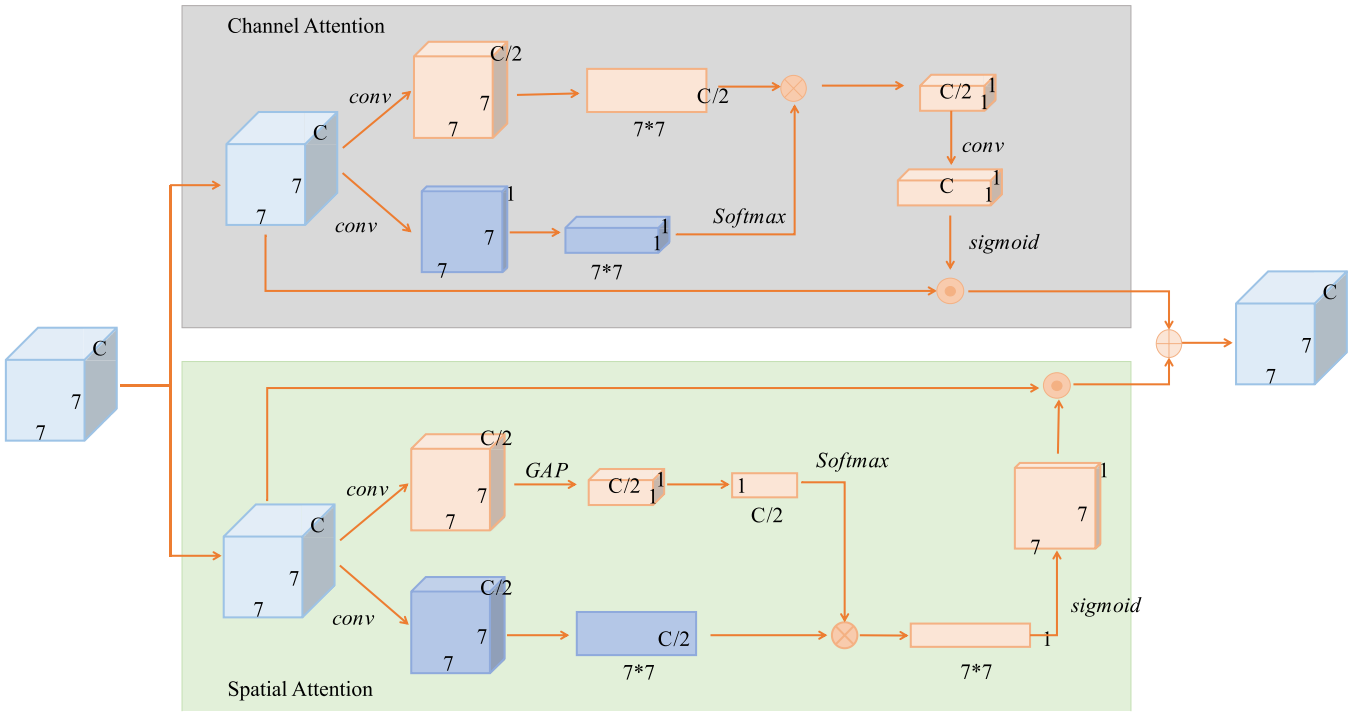


Fig. 5. Hybrid attention module.

novel hybrid channel and spatial attention module to extract the discriminative features of the target.

Channel Attention: Given the input feature x. First, use two convolution kernels with a size of 1*1 to compress the input feature channel into $1 \times H \times W$ and $C/2 \times H \times W$, denoted as f_c^q and f_c^k, respectively. Second, f_c^q performs attention weighted mapping on f_c^k to the key vector, denoted as f_c^v. Finally, the channel is extended to C through a 1*1 convolution operation and multiplied by the given feature x to perform differential weighting on the channel, highlighting the importance between different channels. The process can be expressed as:

$$f_c^q = F_{\text{conv}}^{1*1}(x) \in R^{1*H*W} \tag{4}$$

$$f_c^k = F_{\text{conv}}^{1*1}(x) \in R^{C/2*H*W} \tag{5}$$

$$f_c^v = F_R\left(f_c^k\right) \otimes \text{Softmax}\left[F_R\left(f_c^q\right)\right] \in R^{C/2*1*1} \tag{6}$$

$$x_{\text{cha}} = x \odot \text{sigmoid}\left(F_{\text{conv}}^{1*1}\left(f_c^v\right)\right) \in R^{C*H*W} \tag{7}$$

where F_{conv}^{1*1} represents the 1*1 convolution, F_R represents the reshape function, $\otimes$ means matrix multiplication operation, and $\odot$ means element-wise product operation.

Spatial Attention: Given input feature x. First, use two convolution kernels with a size of 1*1 to compress the input feature channels into the same size $C/2 \times H \times W$, denoted as f_s^q and f_s^k, respectively. Second, spatial feature aggregation is performed on f_s^q through global average pooling operation. After f_s^q and f_s^k reshape, f_s^q obtains weight factor through softmax operation and multiplies it with f_s^k to obtain f_s^v. Finally, the f_s^v size is reshaped to $1*H*W$ through the reshape operation, and the activation value is obtained by multiplying it with the given feature x through sigmoid function. The feature is differential weighted in the spatial dimension to highlight the target suppression background. The process can be expressed as:

$$f_s^q = \text{GAP}\left(F_{\text{conv}}^{1*1}(x)\right) \in R^{C/2*1*1} \tag{8}$$

$$f_s^k = F_{\text{conv}}^{1*1}(x) \in R^{C/2*H*W} \tag{9}$$

$$f_s^v = F_R\left(f_s^k\right) \otimes \text{Softmax}\left[F_R\left(f_s^q\right)\right] \in R^{1*HW} \tag{10}$$

$$x_{\text{spa}} = x \odot \text{sigmoid}\left(F_R\left(f_s^v\right)\right) \in R^{C*H*W} \tag{11}$$

where GAP represents global average pooling.

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Finally, we will add channel attention and spatial attention to further improve the feature representation:

$$x_{\text{att}} = x_{\text{cha}} + x_{\text{spa}}. \tag{12}$$

To further enhance the feature representation of the boundary, we set the value of the middle 4 pixels of the feature to 0 and only retain the boundary information. Finally, we weigh the attention features and boundary features to obtain a boundary-enhanced feature representation:

$$x = x_{\text{att}} + x_{\text{ba}} \tag{13}$$

where x_{ba} is the boundary feature. Finally, the adaptive representation of spatial perception is obtained by adding, which is used for accurate regression prediction of the target.

D. Label Assignment Based on Gaussian Distribution

The anchor box-based detector determines positive and negative samples through the IoU threshold. For small targets, the ground truth and most anchor boxes overlap to a low degree, which easily causes small targets to miss detection during target regression. Therefore, we introduce the Gaussian distribution to measure the similarity between two boxes in the target detector [39]. The Gaussian distribution can still provide learning information when the two boxes do not overlap, as shown in Fig. 6. Therefore, small targets that do not match positive samples based on IoU rules can match positive samples based on the Gaussian distribution.

In the first stage, FADet presets a series of horizontal anchors with multiple scales on the image. The ground truth is converted into a circumscribed minimum rectangular box, and then the ground truth and anchor are converted into a Gaussian distribution.

Specifically, for the minimum circumscribed HBB of the ground truth $R = (cx, cy, w, h)$, where cx, cy, w, and h represent the center coordinates, width, and height, respectively, the equation of its inscribed ellipse can be expressed as

$$\frac{(x-\mu_x)^2}{\sigma_x^2} + \frac{(y-\mu_y)^2}{\sigma_y^2} = 1 \tag{14}$$

where (μ_x, μ_y) is the center coordinate of the ellipse, and (σ_x, σ_y) is the length of the semiaxes along the x-axis and the y-axis. Therefore, $\mu_x = cx$, $\mu_y = cy$, $\sigma_x = (w/2)$, and $\sigma_y = (h/2)$.

The ellipse in (14) represents the distribution density function of the 2-D Gaussian distribution of a box. Therefore, the HBB $R = (cx, cy, w, h)$ can be modeled as a 2-D Gaussian distribution $N(\mu, \sum)$

$$\mu = \begin{bmatrix} cx \\ cy \end{bmatrix}, \quad \sum = \begin{bmatrix} \frac{w^2}{4} & 0 \\ 0 & \frac{h^2}{4} \end{bmatrix} \tag{15}$$

where μ is the mean and Σ is the covariance matrix. Using the covariance matrix, we can calculate the area of the Gaussian distribution bounding box:

$$V = 2^n \cdot \left|\sum\right|^{\frac{1}{2}}, \quad n = 2. \tag{16}$$

The n represents the dimension.

Convert the anchor and groundtruth box to Gaussian distribution $n_a = N(\mu_a, \Sigma_a)$ and $n_g = N(\mu_g, \Sigma_g)$, respectively. Using the Kalman filtering method to calculate the overlapping area between the anchor box and the truth box in two 2-D Gaussian distributions [40], specifically:

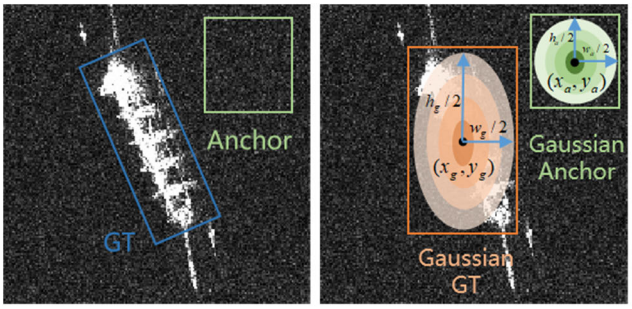


Fig. 6. (Left) GT box and anchor. (Right) GT box and anchor converted into Gaussian distribution. As shown in figure, coordinates of GT frame are $(x_g, y_g, (w_g/2), (h_g/2)) = (80, 80, 20, 40)$, the coordinates of the anchor are $(x_a, y_a, (w_a/2), (h_a/2)) = (120, 120, 20, 20)$, the IoU of GT and the anchor is 0, and KLD distance between Gaussian GT and the Gaussian anchor is approximately 2.8. Thus, two disjoint boxes under Gaussian distribution can still learn to regress information.

$$N\left(\mu, \sum\right) = N\left(\mu_a, \sum_a\right) \cdot N\left(\mu_g, \sum_g\right) \tag{17}$$

where $\mu = \mu_t + K(\mu_a - \mu_a)$, $\Sigma = \Sigma_t - K\Sigma_t$. K represents Kalman gain $K = \Sigma_t(\Sigma_a + \Sigma_t)^{-1}$.

The covariance matrix of overlapping regions can be expressed as:

$$\sum = \sum_t - \sum_t \left(\sum_a + \sum_t\right)^{-1} \sum_t. \tag{18}$$

According to (16), the area of the overlapping area can be determined.

The intersection and union ratio between two boxes after converting to Gaussian distribution is:

$$\text{GsIoU} = \frac{V}{V_a + V_t - V} = \frac{4 \cdot \left|\sum\right|^{\frac{1}{2}}}{4 \cdot \left|\sum_a\right|^{\frac{1}{2}} + 4 \cdot \left|\sum_t\right|^{\frac{1}{2}} - 4 \cdot \left|\sum\right|^{\frac{1}{2}}}. \tag{19}$$

We use the Kullback–Leibler distance to calculate the distribution distance. Kullback–Leibler is a classical calculation of the distribution distance that measures the different degrees of a probability distribution [40]. The KLD distance between n_a and n_g is as follows:

$$D_{\text{KL}}(n_a \parallel n_g) = \frac{1}{2}\left(\text{tr}\left(\sum_g^{-1}\sum_a\right) + (\mu_g - \mu_a)^{\text{T}} \times \sum_g^{-1}(\mu_g - \mu_a) + \ln\frac{\left|\sum_g\right|}{\left|\sum_a\right|} - 2\right). \tag{20}$$

The formula can be further simplified as follows:

$$D_{\text{KL}}(n_a \parallel n_g) = \frac{w_a^2}{2w_g^2} + \frac{h_a^2}{2h_g^2} + \frac{2(x_a - x_g)^2}{w_g^2} + \frac{2(y_a - y_g)^2}{h_g^2} + \ln\frac{w_g}{w_a} + \ln\frac{h_g}{h_a} - 1. \tag{21}$$

Observing formula (21), it can be seen that the KLD distance is related to the ground truth, the center point coordinates

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TABLE I
ABLATION EXPERIMENT

	Oriented RCNN	FAM	BA	GLA	SSDD+							RSDD-SAR							FPS	Params (M)
					P	R	AP50	AP75	AP_s	AP_m	AP_l	P	R	AP50	AP75	AP_s	AP_m	AP_l		
(1)	✓				93.64	93.01	89.58	58.12	44.02	57.65	67.00	90.86	91.43	88.72	57.43	43.82	56.64	65.25	**26.60**	**41.35**
(2)	✓	✓			93.95	93.42	90.08	58.93	44.89	58.13	67.12	93.42	92.79	89.15	58.11	44.67	57.12	65.79	23.18	41.72
(3)	✓	✓	✓		**95.33**	94.64	90.54	59.74	45.08	58.47	67.53	**94.51**	93.36	89.78	58.62	44.78	57.53	66.33	22.24	42.78
(4)	✓	✓	✓	✓	95.51	**94.79**	**91.03**	**59.94**	**46.14**	**58.78**	**67.87**	94.33	**94.72**	**90.78**	**59.91**	**45.82**	**58.17**	**66.99**	19.83	42.78

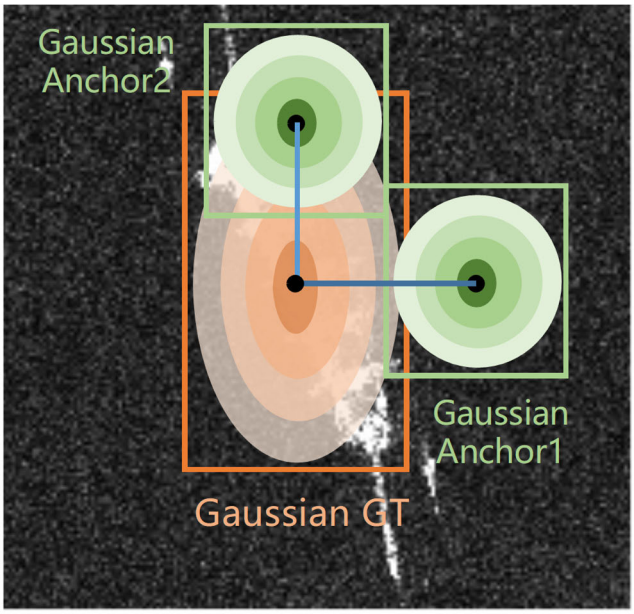


Fig. 7. Two anchors of the same size, with the same center distance and KLD distance from the groundtruth box.

of the anchor box, and the length and width of the anchor box. When the center point coordinates and length and width of the ground truth are known, the KLD distance is only related to the center point position and length and width of the anchor. As shown in Fig. 7, two anchor boxes of the same size have the same distance from the center point to the ground truth, resulting in the same KLD. However, the intersection and union ratio of the two anchor boxes and the ground truth are not the same, and the difficulty of model prediction regression is also different. Therefore, only using the KLD distance cannot accurately measure the Gaussian distribution similarity between two boxes. Based on this, we have designed a new measurement measure as follows:

$$\mathrm{GKL} = \frac{1}{1 + (1 + \alpha \mathrm{GsIoU})\mathrm{KLD}}. \tag{22}$$

Observing (19), it can be seen that the calculation of GsIoU is only related to the length and width of the box, which means that as long as the length and width of the box are fixed, the calculated intersection area will be fixed. Therefore, in order to calculate the true intersection ratio need to satisfy condition $|x_a - x_g| - (w_a + w_g) < 0$. On the basis of KLD, a scaling factor with intersection to union ratio as the coefficient is introduced, when $|x_a - x_g| - (w_a + w_g) < 0$, $\alpha = 0$; otherwise $\alpha = 1$. When KLD is the same, the larger the value of GsIoU, the smaller the GKL, and the greater the similarity between the two Gaussian distributions, that is, the higher the degree of matching between the two boxes.

Before allocation, we calculate the matching score between the ground truth and anchor box based on the above method and sort the scores. Then, the three positive samples are assigned to the corresponding ground truth. To improve the recall rate of small targets, we scale all anchor boxes to 0.9 times the original size and repeat the above sorting strategy to add a positive sample for each ground truth.

IV. EXPERIMENTS

To evaluate the performance of our proposed algorithm, we conducted a series of verification experiments. First, we introduce the datasets and experimental settings. Second, we introduce the evaluation metrics for the experiments. Finally, we verify the effectiveness of each module of our proposed method through ablation experiments. To further verify the effectiveness of our method, we compare FADet with other state-of-the-art optical-based, SAR-based oriented object detectors, and traditional CFAR target detectors.

A. Datasets and Experimental Setup

For the objectivity of the experiment, we used the SSDD+ [41] and RSDD-SAR [42] datasets to verify our experiment. Among them, SSDD+ contains 1160 images, and RSDD-SAR contains 7000 images. The amount of data in the two datasets is quite different, so we can better evaluate the generalization performance of the algorithm.

SSDD+: On the basis of SSDD, Li et al. [41] marked the horizontal box as an oriented box. As the earliest open-source SAR image dataset, SSDD+ has made outstanding contributions to the development of oriented detectors. SSDD+ contains 2456 ship targets, and the average number of ships in each image is 2.12. The images in SSDD+ are obtained by RadarSat-2, TerraSAR-X, and Sendin-1 sensors. The images adopt four polarization modes: HH, HV, VV, and VH. The image size is 500–600, and the resolution is 1–15 m.

RSDD-SAR: RSDD-SAR [42] is a dataset created by Xu et al. in 2022 to solve the problem of a low number of SAR orientation datasets. The RSDD-SAR contains 7000 images obtained by Gaofen-3 and TerraSAR-X satellites. The size of the images is 512*512. There are two polarization modes, HH and HV, and the resolution is 2–20 m. The data include 10 263 ship instance targets and 6146 small targets (fewer than 625 pixels in the area), accounting for 59.88%. Therefore, this is a very challenging dataset.

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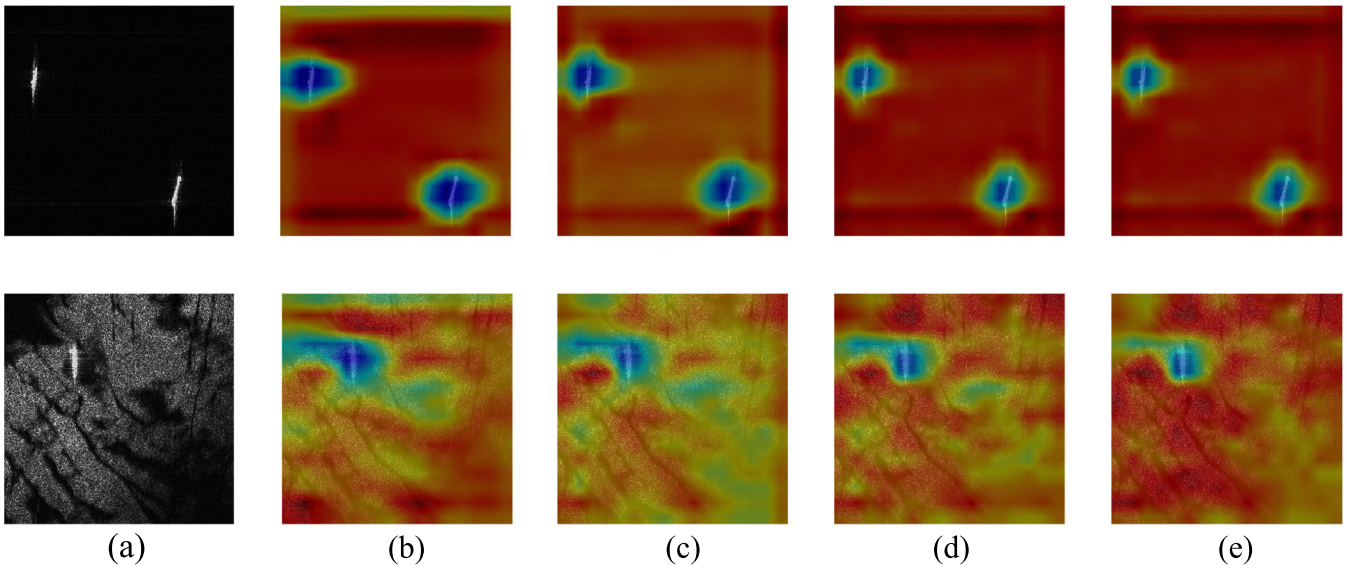


Fig. 8. Feature visualization: (a) is the original image and (b)–(e) represents the visualization results of four ablation experiments (1)–(4) in Table I, respectively.

TABLE II
COMPARISON WITH STATE-OF-THE-ART ORIENTED TARGET DETECTION METHODS

Method		SSDD+			RSDD-SAR			FPS	Params(M)
		P	R	mAP	P	R	mAP		
One stage	R-RetinaNet	76.18	76.53	70.83	72.57	72.31	66.66	22.31	32.44
	R3Det	85.22	86.00	82.72	86.45	88.36	80.87	**29.30**	41.81
	S2ANet	90.38	91.09	88.36	90.95	91.64	87.91	23.46	36.45
	KLD	91.57	91.63	89.10	91.28	92.91	88.32	22.01	37.38
	KFIoU	91.68	93.12	90.25	91.33	92.46	88.41	20.38	37.67
Two stage	RFaster-RCNN	85.64	88.12	84.65	84.98	87.65	83.29	15.87	41.41
	RoI Transformer	90.05	91.31	89.04	89.72	90.53	88.39	11.93	55.32
	Gliding Vertex	88.00	89.62	87.97	88.31	89.46	87.55	16.82	41.41
	ReDet	89.36	92.09	88.65	90.13	90.24	88.40	15.90	**31.57**
	FADet	**94.51**	**94.79**	**91.03**	**94.33**	**94.72**	**90.78**	19.83	42.78

All experiments in this article were conducted on a workstation with an Ubuntu 18.04 system and a GeForce GTX2080Ti. For the convenience of comparative experiments, we randomly divide the SSDD+ datasets into an 8:2 training set and a test set. For RSDD-SAR, we use the division method in [42] with 5000 images as the training set and 2000 images as the test set. The input image size is uniformly scaled to 512*512. Only the RRandomFlip online enhancement strategy is used for training, and the batch size is set to 2. The optimizer uses SGD. The initial learning rate is set to 0.005, the momentum is 0.9, and the weight decay is 0.0001. A total of 100 epochs are trained.

B. Evaluation Index

To comprehensively evaluate the performance of FADet, we conducted a series of experiments in terms of accuracy and speed. In terms of precision, performance indicators such as precision (P), recall (R), and average precision (mAP) are used. In terms of speed, model parameters and FPS indicators are selected. P and R are calculated as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}. \tag{23}$$

TP is true positive and FP is false positive.

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{24}$$

FN is false negative.

The mAP represents the average value of each category of AP, which is the area enclosed by the PR curve. In this article, we use the 11-point method to calculate the PR area. When there is only one class, the mAP is equivalent to AP, which is

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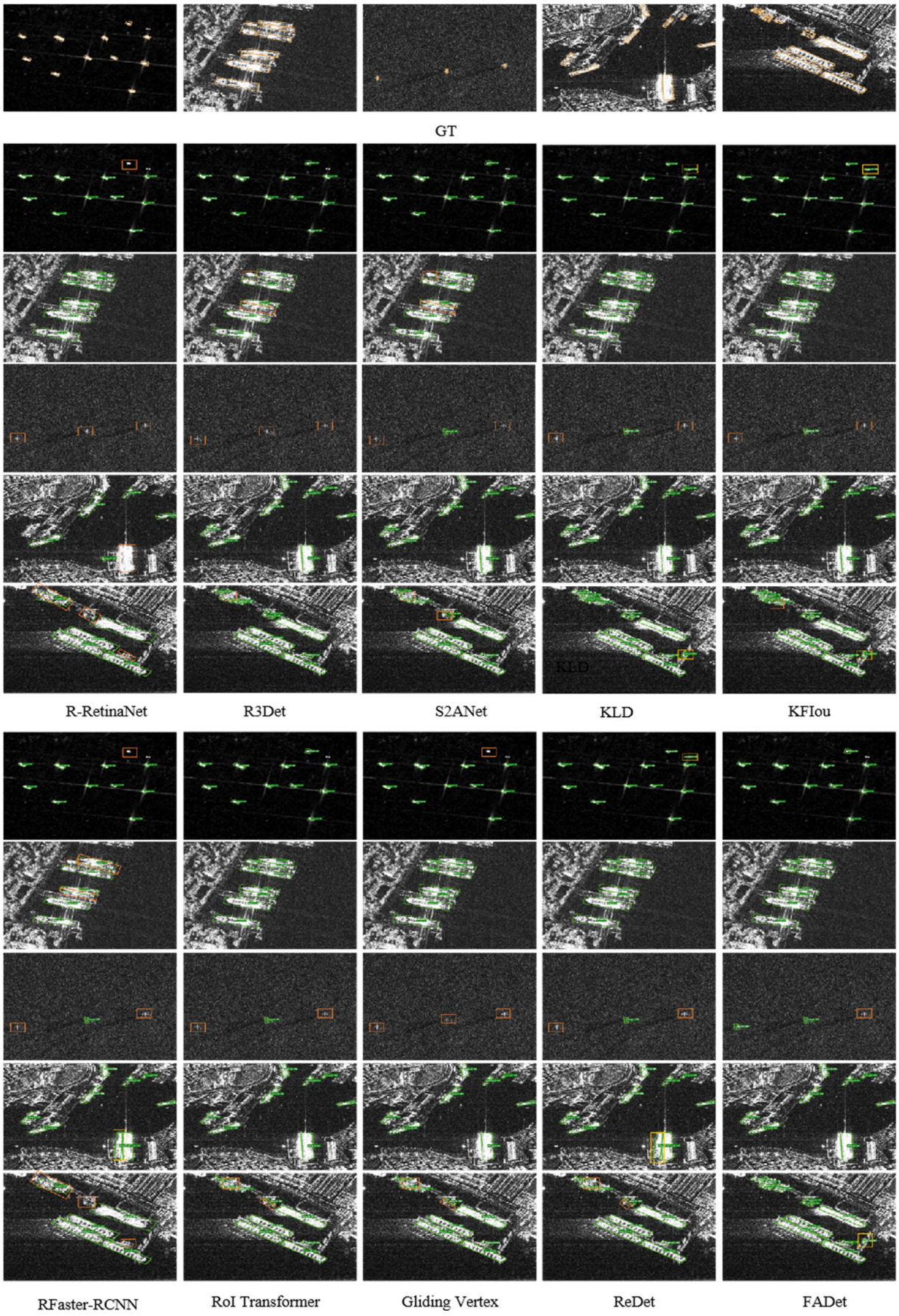


Fig. 9. Visualization of oriented detection algorithm in SSDD+ and RSDD-SAR dataset.

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defined as follows:

$$\text{mAP} = \frac{1}{11} \sum_{r \in S} P|_{R=0,0.1,0.2,\ldots,1}. \tag{25}$$

FPS indicates the number of images that can be processed by the model per second and the real-time detectability of the FPS test algorithm.

C. Experimental Results and Analysis

1) Ablation Experiment: To verify the effectiveness of each module of our method, we conducted four ablation experiments on the SSDD+ and RSDD-SAR datasets and fully verified the accuracy, recall, AP50, AP75, AP_s, AP_m, AP_l model parameters, and FPS indicators of each experiment to comprehensively evaluate the effectiveness of each module. Among them, AP_s, AP_m, AP_l three indicators use IoU from 0.5 to 0.95, with an interval of 0.05 to calculate mAP. The average mAP is taken as the final result.

1) *Oriented RCNN:* As shown in Table I, the AP50, AP75, AP_s, AP_m, AP_l of the oriented RCNN on SSDD+ respectively is 89.58, 58.12, 44.02, 57.65, and 67.00. On RSDD-SAR is 88.72, 57.43, 43.82, 56.64, and 65.25. In the four ablation experiments, the detection accuracy of the oriented RCNN is the lowest, but the FPS of the oriented RCNN is 26.60, the reasoning speed is the fastest, and the model parameters are the lowest.
2) *FAM:* The AP50 and AP75 of FAM on SSDD+ and RSDD-SAR are 90.08, 58.93 and 89.15, 58.11, respectively. AP50 are 0.5 and 0.43 higher than the benchmark oriented RCNN, AP75 are 0.71 and 0.68 higher, it indicates that under stricter AP75 indicators, the performance increases even higher. The FPS is reduced by 3.42, and the number of parameters is increased by only 0.37. The reason for this may be that the warp in the FAM consumes a certain amount of time.
3) *BA:* According to the experimental results, the accuracy of the added BA module on SSDD+ is the highest at 95.33. The AP50, AP75, AP_s, AP_m, AP_l on SSDD+ are 90.54, 59.74, 45.08, 58.47, 67.53 and on RSDD-SAR are 89.78, 58.62, 44.78, 57.53, 66.33, respectively. Compared with the benchmark algorithm of plus FAM, AP50 increases by 0.46 and 0.63, AP75 increases by 0.84 and 1.73, AP_s increases by 0.19 and 0.11, AP_m increases by 0.36 and 0.41, AP_l increases by 0.41 and 0.54. FPS decreases by 1.16. As a stricter evaluation indicator, AP75 has a significant increase in detection accuracy, indicating that the addition of boundary adaptive modules has better detection performance when IoU is equal to 0.75.
4) *GLA:* As shown in Table I, FADet achieves the highest precision in terms of recall (94.79), AP50 (91.03), AP75 (59.94), AP_s (46.14), AP_m (58.78), and AP_l (67.87) on the SSDD+ datasets. The RSDD-SAR datasets have the highest recall (94.72), AP50 (90.78), AP75 (59.91), AP_s (45.82), AP_m (58.17), and AP_l (66.99), respectively. Compared to Experiment (3), the performance of small targets has increased 1.06 on SSDD+ and 1.04 on RSDD-SAR. This demonstrates that using the GLA matching rule can effectively increase the performance of small target detection.

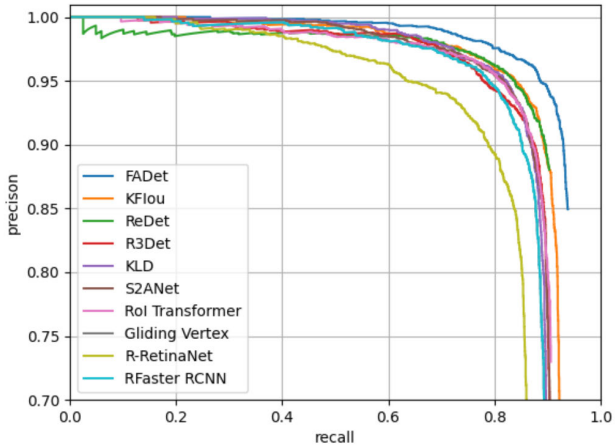


Fig. 10. PR curves of state-of-the-art directed object detectors on the RSDD-SAR dataset.

TABLE III
COMPARISON OF ORIENTED SAR TARGET DETECTION ALGORITHMS BASED ON DEEP LEARNING ON SSDD+

Method	SSDD+		
	P	R	mAP
DRBox-v2	/	/	92.81
R2FA-Det	/	/	94.74
OFF-Net	93.93	92.07	95.05
SKGNet	92.58	94.33	95.02
MSR2N	86.52	92.05	90.11
FBR-Net	92.79	94.01	94.10
FADet	**95.47**	**95.93**	**95.81**

TABLE IV
COMPARED WITH TRADITIONAL SAR TARGET DETECTION ALGORITHMS ON SSDD+

Method	SSDD+		
	P	R	mAP
CA-CFAR [52]	86.41	93.56	87.25
OS-CFAR [53]	87.12	92.49	87.64
Multi-CFAR [54]	75.39	90.11	73.28
Two-Parameter CFAR [55]	84.45	94.18	84.77
FADet	**95.47**	**95.93**	**95.81**

According to the ablation experiment, the detection performance of our FADet is more than 1.5% higher than that of the benchmark model on both datasets. Additionally, we can see that the model speed decreases by 6.77. We hypothesize that

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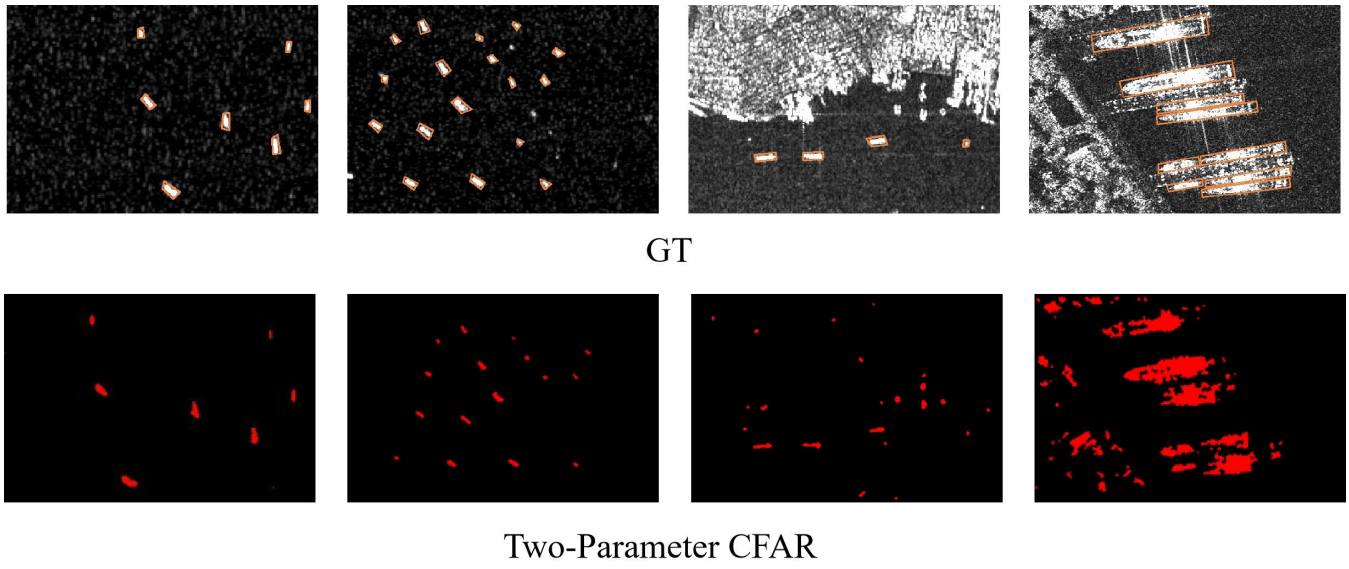


Fig. 11. GT and two-parameter CFAR detection results.

the reason may be that the semantic flow alignment process consumes a certain amount of time. In summary, our method has slightly reduced the speed on the premise of considerably improving the accuracy.

To further understand the decision-making mechanism of the method, we performed a visual analysis of the characteristics of four ablation experiments. To be more comprehensive, we selected a feature of the fourth stage of the visualization backbone network of inshore and offshore targets. Fig. 8(b)–(e) represent the visualization results of the four ablation experiments. The greater the blue area is concentrated around the target, the greater the model has learned the target information. It can be seen from Fig. 8(e) that the blue area of the FADet is more concentrated around the target, indicating that the weights learned by FADet pay more attention to the target information and suppress the background information. Thus, the classification and positioning effect is excellent.

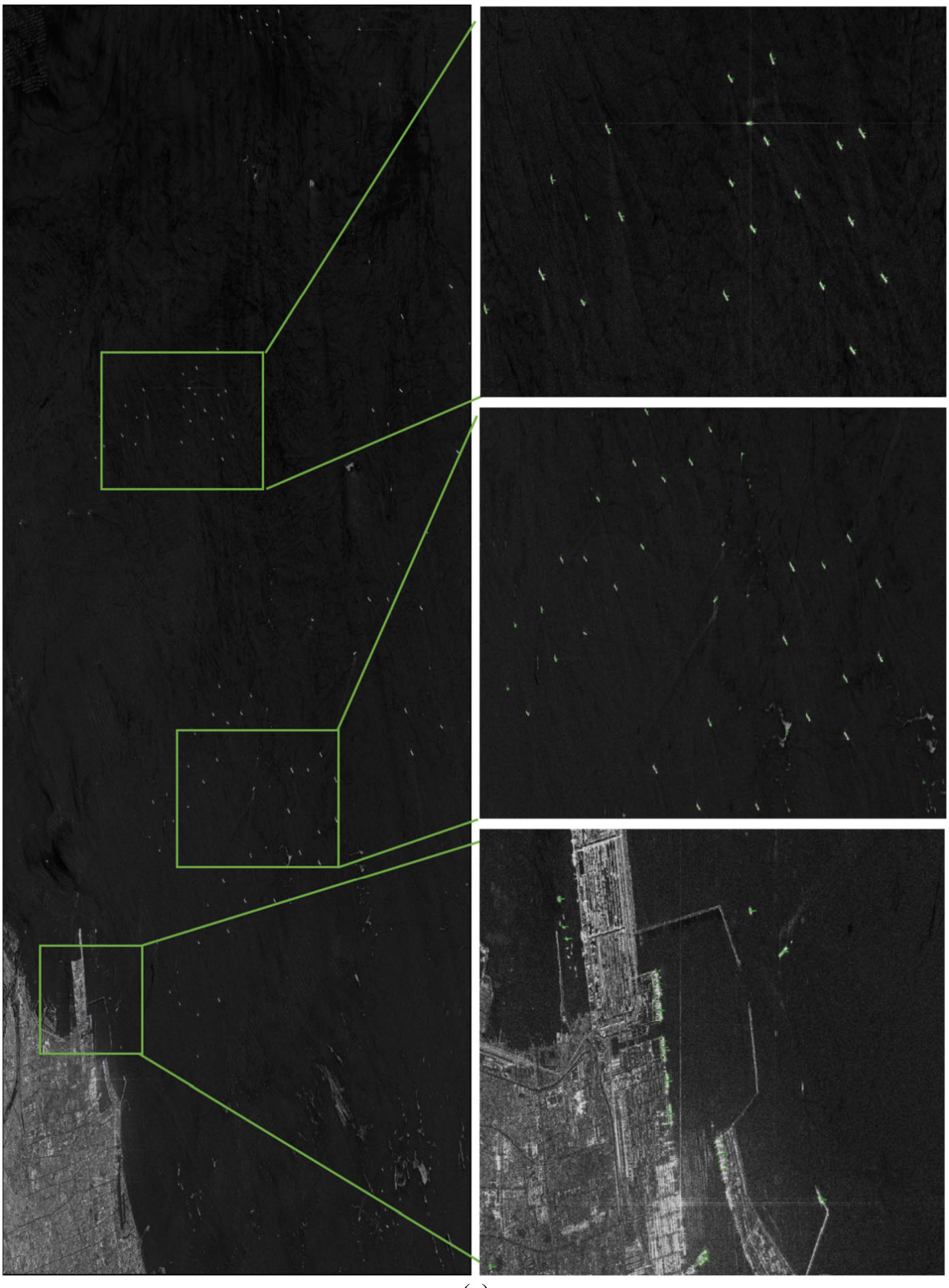
2) Comparison With State-of-the-Art Algorithms: To further verify the effectiveness of FADet, we performed performance comparison experiments with nine CNN-based methods and visualized several images to intuitively observe the prediction results. In addition, we printed the PR curve of each algorithm on RSDD-SAR.

ResNet50 as the backbone, has strong feature extraction capability. Most algorithms use ResNet50 as a feature extraction network to ensure the balance of speed and accuracy. To highlight the impact of the overall algorithm on the performance, it is necessary to ensure that the computational complexity of the backbone network is at the same level. In Table II, except for ReDet, which uses ReResNet50, we uniformly choose ResNet50 as the feature extraction network. In the one-stage algorithm, we compared five algorithms: R-RetinaNet, R3Det, S2ANet, KLD, and KFIoU. KFIoU has the highest performance. The mAP on SSDD+ and RSDD-SAR are 90.25 and 88.41, respectively. However, the inference speed is relatively slow, and the FPS is 20.38. Then, we compared four two-stage algorithms: RFaster RCNN, RoI Transformer, gliding vertex [43], and ReDet. The best performance on SSDD+ is RoI Transformer with an mAP of 89.04. The best performance on the RSDD-SAR datasets is ReDet with an mAP of 88.40. The fastest inference speed is gliding vertex, with an FPS of 16.82. Compared with the KFIoU on SSDD+ and RSDD-SAR, the mAP of our FADet increases by 0.78 and 2.37, respectively. Compared with the RoI transformer with the best performance on SSDD+, the mAP increases by 1.99. For RSDD-SAR and the best performance of ReDet, mAP increases by 2.38. Our method has the fastest inference speed among the two-stage algorithms, which shows the effectiveness of our algorithm in terms of accuracy and speed.

In addition, we visualized the detection results of each algorithm. The first two lines of each algorithm selected images from SSDD+, and the last three lines selected images from RSDD-SAR. As shown in Fig. 9, we selected several representative offshore and inshore targets to test the algorithm performance. The green box in the image represents the detection box, the orange box represents the undetected target, and the yellow box represents the false alarm. It can be seen from the third line of each algorithm in Fig. 9 that most algorithms do not detect targets or only detect one target, indicating the difficulty of small target detection. Our proposed FADet algorithm detects two targets, indicating the effectiveness of label allocation based on a Gaussian distribution. In addition, according to the visualization results of images under complex backgrounds, most algorithms have missed detections and false alarms, which indicates that the algorithm has limited ability to highlight target information and suppress background information, leading to detection errors. The FADet has only one false alarm detection under a complex background, which reflects the robustness of the algorithm. Fig. 10 shows the PR curve on the RSDD-SAR dataset and further verifies the effectiveness of FADet.

3) Comparison With the SAR Target Detection Algorithms Based on Deep Learning: To further verify the performance of FADet, we performed further comparison experiments with the most advanced oriented detection algorithm based on SAR images. We compared six methods: DRBox-v2 [32], R2FA-Det [35], OFF-Net [37], SKGNet [44], MSR2N [45], and FBR-Net [46]. Three indicators were used, precision,

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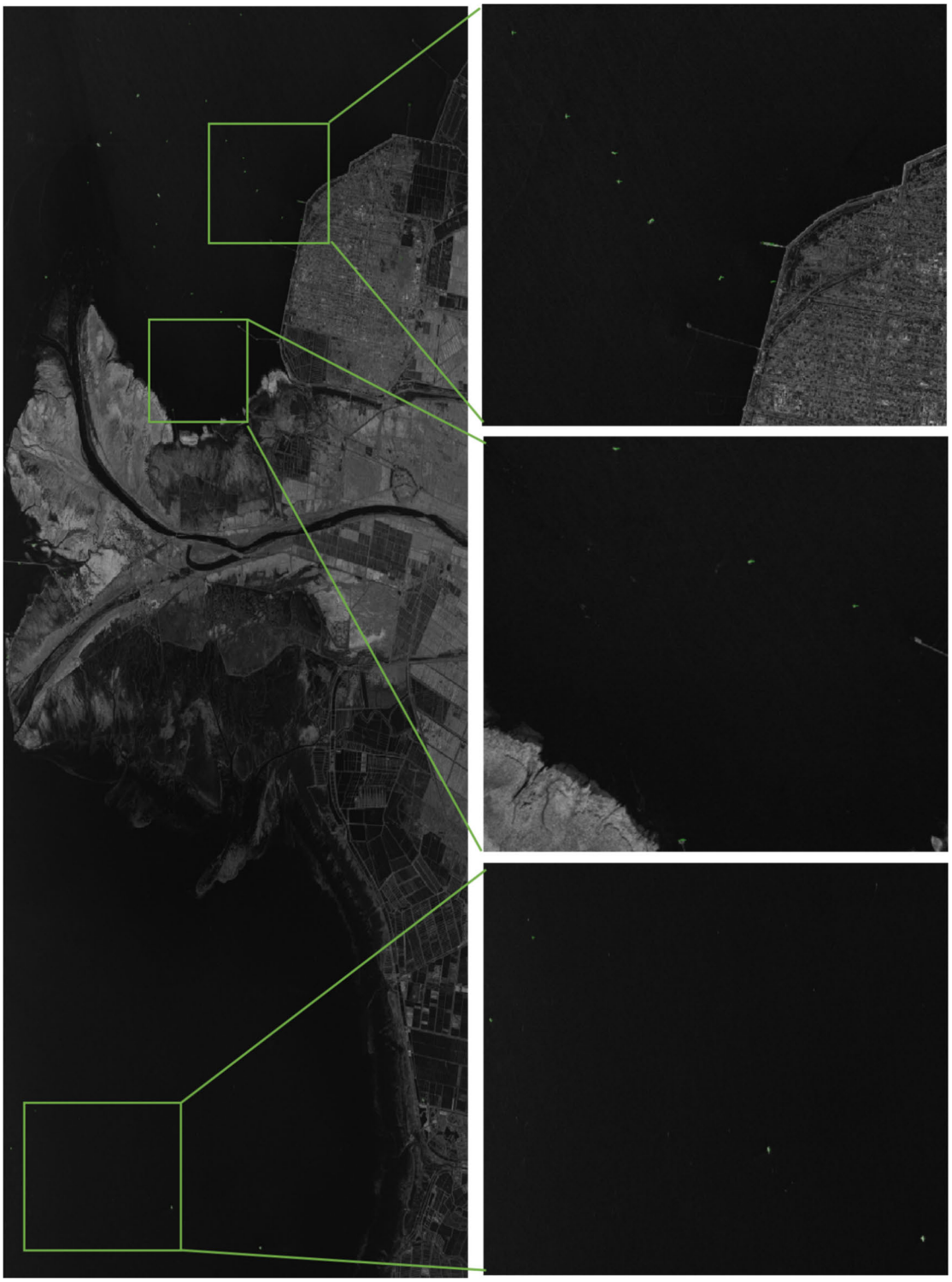


(a)

Fig. 12. (a) Visualization of FADet algorithm on large scene images.

recall, and mAP. These six methods use the area surrounded by the PR curve to calculate mAP. To maintain comparability, in this section, we use (25) to calculate mAP. As shown in Table III, the precision, recall, and mAP of FADet on the SSDD+ datasets are 95.47, 95.84, and 95.81, respectively. The precision is 1.44 higher than 93.93 of the highest OFF-Net, the recall is 1.92 higher than 94.01 of the highest FBR-Net, and the mAP is 0.59 higher than 95.02 of the highest OFF-Net. From these three indicators, FADet has the highest performance, reflecting

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(b)

Fig. 12. *(Continued.)* (b) Visualization of FADet algorithm on large scene images.

the effectiveness of our proposed method.

$$\text{mAP} = \int_0^1 P(R)dR. \tag{26}$$

4) Compared With Traditional SAR Target Detection Algorithms: In order to further verify the performance of FADet, we compared several classic CFAR algorithms in this section. Due to the different implementation methods of the CFAR algorithm and deep learning algorithm, the CFAR algorithm does not require a training set to train the model. Therefore, all experiments in this section were conducted on the test set, and the detection accuracy, recall, and mAP were ultimately calculated based on the detection results. As shown in Table IV, the OS-CFAR with the best performance has the highest mAP of 87.64, which is much lower than the 95.81 of our proposed FADet. During the experiment, we found that the detector

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based on the CFAR algorithm performs better in detecting offshore and small targets, but performs worse in detecting inshore targets, as shown in Fig. 11, many background clutter were detected in inshore targets, which greatly affected the detection results of nearshore targets. The reason is that CFAR algorithm is a pixel level object detection algorithm that achieves object detection by determining whether the grayscale value of each pixel exceeds a preset value. The background of inshore targets is complex, and the statistical model of clutter affects the performance of the detector. However, the CFAR detector has good detection performance and fast speed for offshore targets.

5) Verification of Algorithm Performance on Large Scene Images: In practical applications, SAR images are generally large scene images. To better verify the practicability of the algorithm, we performed a visualization experiment of the detection results of the FADet on two large images provided by RSDD-SAR. As shown in Fig. 12(a) and (b), for each image, we enlarged three areas with dense ship targets. It can be seen from the Fig. 12 that most ship targets can be detected, especially offshore targets. The detection effect is good, indicating that our algorithm has good robustness and can obtain good detection results on data in different scenes.

V. Conclusion

In this article, a new oriented SAR ship target detection algorithm is proposed. Based on the oriented R-CNN, an FAM module is introduced into the FPN to solve the feature dislocation problem caused by downsampling residual connections between deep and shallow features through semantic flow. Second, an adaptive boundary enhancement module is proposed that extracts salient information through the attention mechanism and performs an addition operation to enhance the boundary information of the target, thereby improving the localization performance of the target. Finally, a label allocation strategy based on a Gaussian distribution is proposed to resolve the problem that small targets cannot match positive samples based on IoU matching rules, which leads to small target misdetection. Through ablation experiments on the SSDD+ and RSDD-SAR datasets, the effectiveness of each module of our algorithm is demonstrated. According to a comparison experiment of the latest oriented target detection methods, the AP50 of FADet on the two datasets was 91.03 and 90.78, which shows the progressiveness of our proposed algorithm. Visualization experiments show that our algorithm is robust to small targets and complex backgrounds.

Most SAR target detection algorithms were migrated from optical image detection algorithms. The imaging principles of SAR images and optical images are essentially different. The unique characteristics of SAR images have not been applied to an algorithm to guide algorithm learning. Therefore, in the future, we will apply the SAR imaging mechanism to the detection algorithm to improve the interpretability of the algorithm.

In addition, target detection in SAR images has difficulty detecting a specific type of target and can only detect the general attributes of the target, which limits the practical application of SAR target detection in high fine-grained detection. Therefore, in the future, we will combine optical images and SAR image detection from the perspective of multiple sources and use a small number of optical images to provide the target's fine-grained features to guide the algorithm to extract the fine-grained features of SAR images.

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